Java Foundation Problems — Algorithms, Logic Flow & Core Concepts 📘

This document organizes the given problems into structured categories with:

- Problem statement
- Step-by-step algorithm
- Key logic
- Concepts used

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1. Sum of Two Numbers

Problem

Accept two integers from the user and print their sum.

Example

Input: 45, 12 Output: The sum of 45 & 12 = 57

Algorithm

1. Start
2. Declare two integer variables `a` and `b`
3. Take input from user
4. Calculate `sum = a + b`
5. Print the result
6. End

Concepts Used

- Variables
- Input/Output
- Arithmetic Operators

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2. Name and Age

Problem

Accept user's name and age and print: Hello Shery, you are 12 years old.

Algorithm

1. Start
2. Declare `String name`
3. Declare `int age`
4. Take input
5. Print formatted message
6. End

Concepts Used

- String
- Variables
- Concatenation
- Input/Output

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3. Increment & Decrement Operators

Concepts

Increment

- Pre Increment: `++a`
- Post Increment: `a++`

Decrement

- Pre Decrement: `--a`
- Post Decrement: `a--`

Key Difference

- Pre → value changes first
- Post → value used first, then changes

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4. Swap Two Numbers

Method 1 — Using Third Variable

Algorithm

1. Store `a` in `temp`
2. Put `b` into `a`
3. Put `temp` into `b`

Method 2 — Without Third Variable

Algorithm

1. `a = a + b`
2. `b = a - b`
3. `a = a - b`

Concepts Used

- Variables
- Arithmetic Operations
- Swapping Logic

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5. Rectangle Area & Perimeter

Formula

Area = length × width Perimeter = 2 × (length + width)

Algorithm

1. Take length and width
2. Calculate area
3. Calculate perimeter
4. Print results

Concepts Used

- Formula Based Problems
- Arithmetic Operators

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6. Compound Interest

Formula

CI = P × (1 + R / 100)^T

Algorithm

1. Input principal, rate, time
2. Use `Math.pow()`
3. Calculate compound interest
4. Print answer

Concepts Used

- Math Class
- Exponents
- Formula Implementation

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7. Area of Triangle Using Heron's Formula

Formula

s = (a + b + c) / 2

Area = √(s(s-a)(s-b)(s-c))

Algorithm

1. Input three sides
2. Calculate semi perimeter
3. Apply Heron formula
4. Print area

Concepts Used

- Math.sqrt()
- Formula Problems

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8. Surface Area of Sphere

Formula

$4 \times \pi \times r^2$

Concepts Used

- Geometry
- Math Formula

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9. Circumference & Area of Circle

Formula

Circumference = $2\pi r$ Area = πr^2

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10. Greatest Between Two Numbers

Algorithm

1. Input two numbers
2. Compare using `if-else`
3. Print greater number

Concepts Used

- Conditional Statements

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11. Greeting Based on Gender

Logic

- M/m → Good Morning Sir
- F/f → Good Morning Ma'am
- Else → Wrong Input

Concepts Used

- Character Input
- if-else ladder
- Logical OR operator

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12. Even or Odd

Logic

If `number % 2 == 0` → Even Else → Odd

Concepts Used

- Modulus Operator
- Conditions

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13. Valid Voter Program

Logic

- Age ≥ 18 → Valid Voter
- Else → Invalid

Part 2

Print years left: `18 - age`

Concepts Used

- Conditions
- Arithmetic

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14. Day Number to Day Name

Logic

1 → Monday 2 → Tuesday ... 7 → Sunday

Concepts Used

- if-else
- switch statement

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15. Greatest Among Three Numbers

Algorithm

1. Compare all three numbers
2. Print maximum

Concepts Used

- Nested if
- Logical Operators

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16. Leap Year

Logic

A year is leap year if:

- divisible by 400 OR

- divisible by 4 but not by 100

Concepts Used

- Logical Conditions
- Modulus Operator

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17. Shop Discount

Common Logic

- Based on amount
- Apply percentage discount

Concepts Used

- Conditions
- Percentage Calculations

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18. Movie Rating Program

Example Logic

- Age restrictions
- Rating based recommendations

Concepts Used

- Decision Making

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19. Profile Based Program

Concepts Used

- Nested Conditions
- User Input Handling

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20. Bijli Bill

Common Logic

Units consumed determine bill amount.

Concepts Used

- Slab Calculation
- Conditions

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21. Consonant or Vowel

Logic

Check character: A, E, I, O, U

Else consonant.

Concepts Used

- Character Handling
- Conditions

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22. Number of Days in Month

Logic

- 31-day months
- 30-day months
- February special case

Concepts Used

- Conditions
- Leap Year

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23. Print Hello World n Times

Algorithm

1. Input n
2. Use loop
3. Print message n times

Concepts Used

- Loops

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24. Print Natural Numbers up to n

Algorithm

Loop from 1 to n.

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25. Reverse Loop

Algorithm

Loop from n to 1.

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26. Multiplication Table

Algorithm

Loop from 1 to 10. Multiply each by number.

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27. Sum up to n Terms

Formula

$n(n+1)/2$

OR Use loop accumulation.

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28. Factorial of Number

Formula

$n! = n \times (n-1) \times \ldots \times 1$

Algorithm

Loop from 1 to n. Multiply result.

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29. Sum of Even & Odd Numbers in Range

Logic

Use `% 2`. Maintain two sums.

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30. Factors of Number

Algorithm

Loop from 1 to n. If `n % i == 0` → factor

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31. Sum of Factors

Algorithm

1. Find all factors
2. Add them

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32. Prime Number

Logic

A prime number has exactly 2 factors.

Optimized Approach

Check divisibility till $\sqrt{n}$.

Concepts Used

- Loops
- Conditions
- Optimization

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33. Power Calculation

Logic

Find `a^b`

Algorithm

Multiply `a` by itself `b` times.

OR Use `Math.pow(a,b)`

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34. Separate Digits

Logic

Use `% 10` and `/ 10`

Example

123 3 2 1

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35. Sum of Digits

Algorithm

1. Extract digit
2. Add to sum
3. Remove digit

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36. Reverse a Number

Formula

reverse = reverse * 10 + digit

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37. Palindrome Number

Logic

Reverse number. Compare with original.

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38. Strong Number

Logic

Sum factorial of each digit. Compare with original.

Example: 145 = 1! + 4! + 5!

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39. Automorphic Number

Logic

Square ends with same number.

Example: 76 → 5776

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40. ISBN Number

Concepts Used

- Digit Processing
- Validation Formula

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41. Harshad Number

Logic

Number divisible by sum of digits.

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42. Perfect Square

Logic

Find square root. If integer → perfect square.

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43. Abundant Number

Logic

Sum of proper divisors > number.

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44. Number with Exactly x Divisors

Algorithm

1. Count divisors
2. Compare with x

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45. Prime Factors

Algorithm

Divide repeatedly by prime numbers.

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46. Neon Number

Logic

Sum of digits of square = original number.

Example: $9^2 = 81$ 8+1 = 9

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47. Armstrong Number

Logic

Sum of powers of digits equals original number.

Example: $153 = 1^3 + 5^3 + 3^3$

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48. Perfect Number

Logic

Sum of proper divisors equals number.

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49. Friendly Pair

Logic

Compare abundancy index.

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50. Permutations

Formula

$nPr = n! / (n-r)!$

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51. Maximum Handshakes

Formula

$n(n-1)/2$

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52. Fibonacci Even Indexed Sum

Problem

Find: F2 + F4 + F6 + ... + F2n

Algorithm

1. Generate Fibonacci series
2. Track even indexed terms
3. Add them

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53. do-while Hello Program

Logic

Repeat until wrong input.

Concepts Used

- do-while loop

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54. Calculator Using do-while

Features

- Addition
- Subtraction
- Multiplication
- Division

Concepts Used

- switch
- loops

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55. Weekday Using Switch

Concepts Used

- switch statement

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56. Consonant/Vowel Using Switch

Concepts Used

- switch-case

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57. Area Calculator Using Switch

Menu

1. Circle
2. Rectangle
3. Triangle

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58. Project — Guess Game

Logic

1. Generate random number
2. User guesses
3. Compare guesses

Concepts Used

- Loops
- Conditions
- Random Class

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59. Project — Restaurant v1.0

Features

- Menu display
- Order system
- Bill generation

Concepts Used

- switch
- loops
- arithmetic

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60. Star & Pattern Problems ⭐

Types

- Right Triangle
- Inverted Triangle
- Mirrored Triangle
- Alphabet Triangle
- X Pattern
- V Pattern

Core Logic

Use nested loops.

Concepts Used

- Nested loops
- Pattern logic

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61. Array Sum & Average

Algorithm

1. Create array of size n
2. Input elements
3. Find sum

4. Average = sum/n

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62. Greatest Element in Array

Algorithm

1. Assume first element maximum
2. Traverse array
3. Update max

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63. Second Greatest Element

Logic

Maintain:

- largest
- second largest

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64. Check Array Sorted or Not

Algorithm

Traverse array. If any previous > next → NOT sorted

Else sorted.

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65. Reverse Copy Array

Algorithm

1. Input array
2. Create second array
3. Copy in reverse order

Important Mathematical Formulas 📐

Circle

Area = πr^2 Circumference = $2\pi r$

Rectangle

Area = $l \times w$ Perimeter = $2(l+w)$

Triangle

Heron Formula: $\sqrt{}(s(s-a)(s-b)(s-c))$

Sphere

Surface Area = $4\pi r^2$

Permutation

$nPr = n!/(n-r)!$

Handshakes

$n(n-1)/2$

Core Programming Concepts Used 🧠

1. Variables

Store data values.

2. Data Types

- int
- double
- char
- String

3. Operators

- Arithmetic
- Relational
- Logical
- Increment/Decrement

4. Conditional Statements

- if
- if-else
- nested if
- switch

5. Loops

- for
- while
- do-while

6. Arrays

Store multiple values.

7. Functions / Methods

Reusable code blocks.

8. Math Class

- Math.pow()
- Math.sqrt()
- Math.random()

9. Pattern Programming

Uses nested loops.

10. Number Theory

- Prime
- Armstrong
- Perfect
- Palindrome
- Harshad

- Neon

11. Time Complexity Basics

- Single loop → O(n)
- Nested loop → $O(n^2)$

Recommended Learning Order 🚀

1. Variables & Input/Output
2. Operators
3. Conditions
4. Loops
5. Nested Loops
6. Number Problems
7. Patterns
8. Arrays
9. Functions
10. Small Projects

Final Advice 📌

To master these problems:

- Dry run every program manually
- Focus on logic before syntax
- Practice loops and conditions heavily
- Learn pattern visualization
- Understand number manipulation deeply
- Solve same problem using multiple methods

These problems collectively build:

- Logical thinking
- Mathematical reasoning
- Problem solving
- Loop mastery
- Condition handling
- Array manipulation foundation